



Logistic regression

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Hierarchy of Study Design (mine)

- Systematic review + meta analysis
- Randomized designs (RCT)
- Natural experiments
- Cohort studies
- Case control studies
- Cross-sectional & ecological studies

Study design and regression

- All study designs can and do use regression to estimate associations (effects)
- It's crucial to consider the design when thinking about regression.
 - An Odds Ratio (OR) from one study is not the same as one from another study.

Regression in general

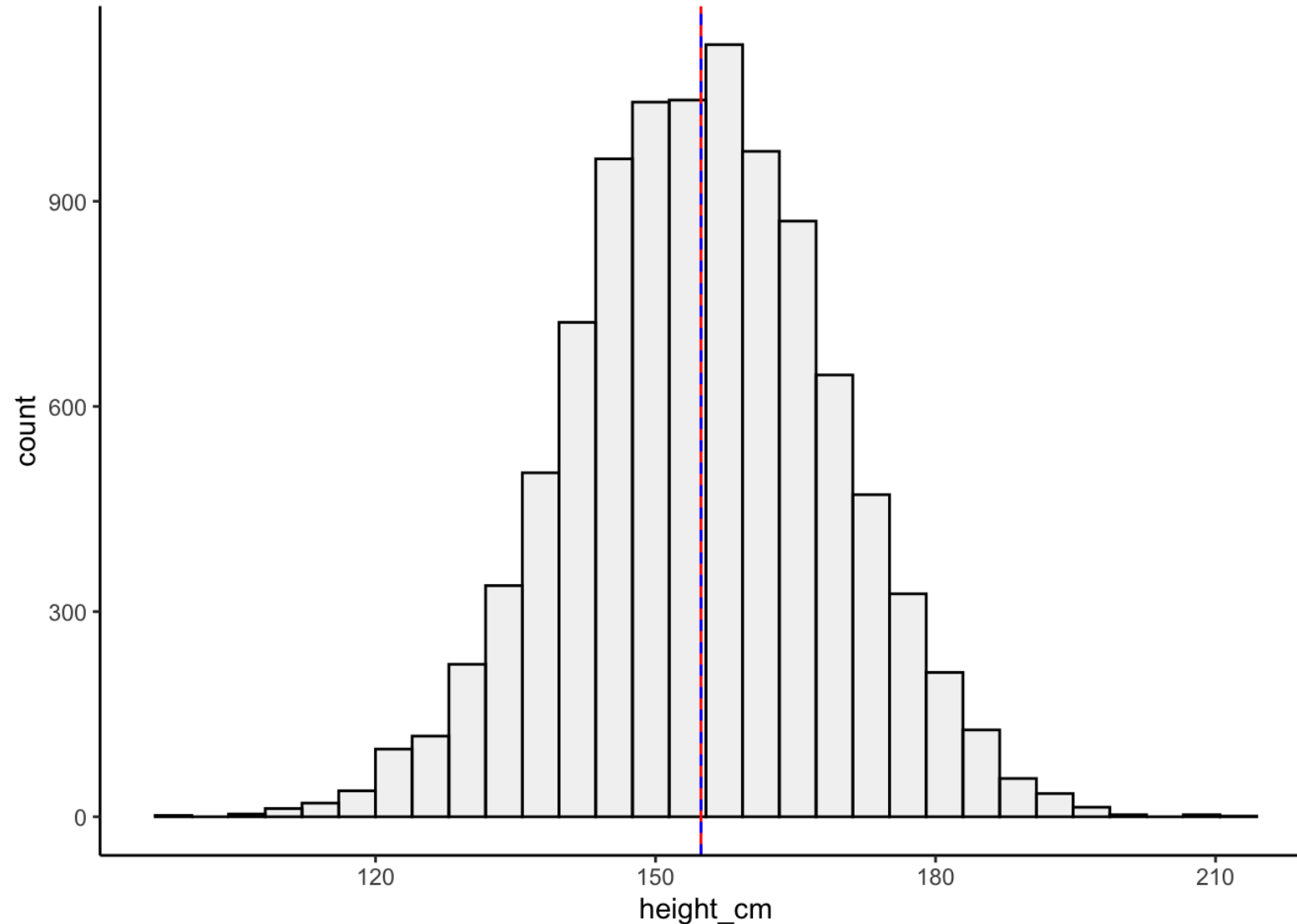
- Create a model that fits the data
- Common model types are named because of the distribution of the outcome variable
 - Or assumptions required for the models distribution

Common Distributions

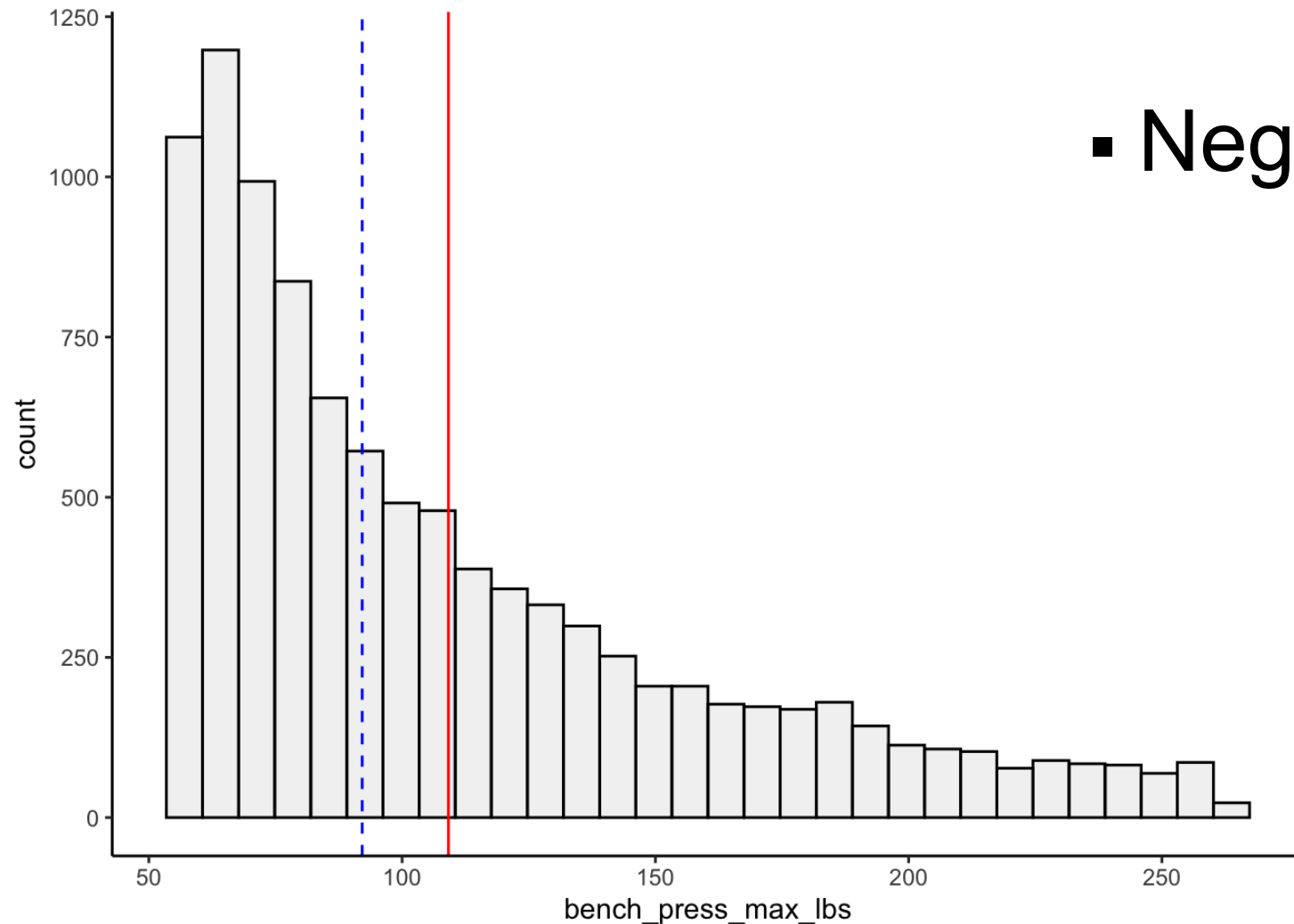
- Normal or Gaussian
 - a) Bell curve
- Uniform
- Count type
 - a) Poisson and negative binomial
- Bernoulli
 - a) Yes or no
- Many, many, more
 - a) https://en.wikipedia.org/wiki/List_of_probability_distributions

Normal (Gaussian) distribution

- Linear regression

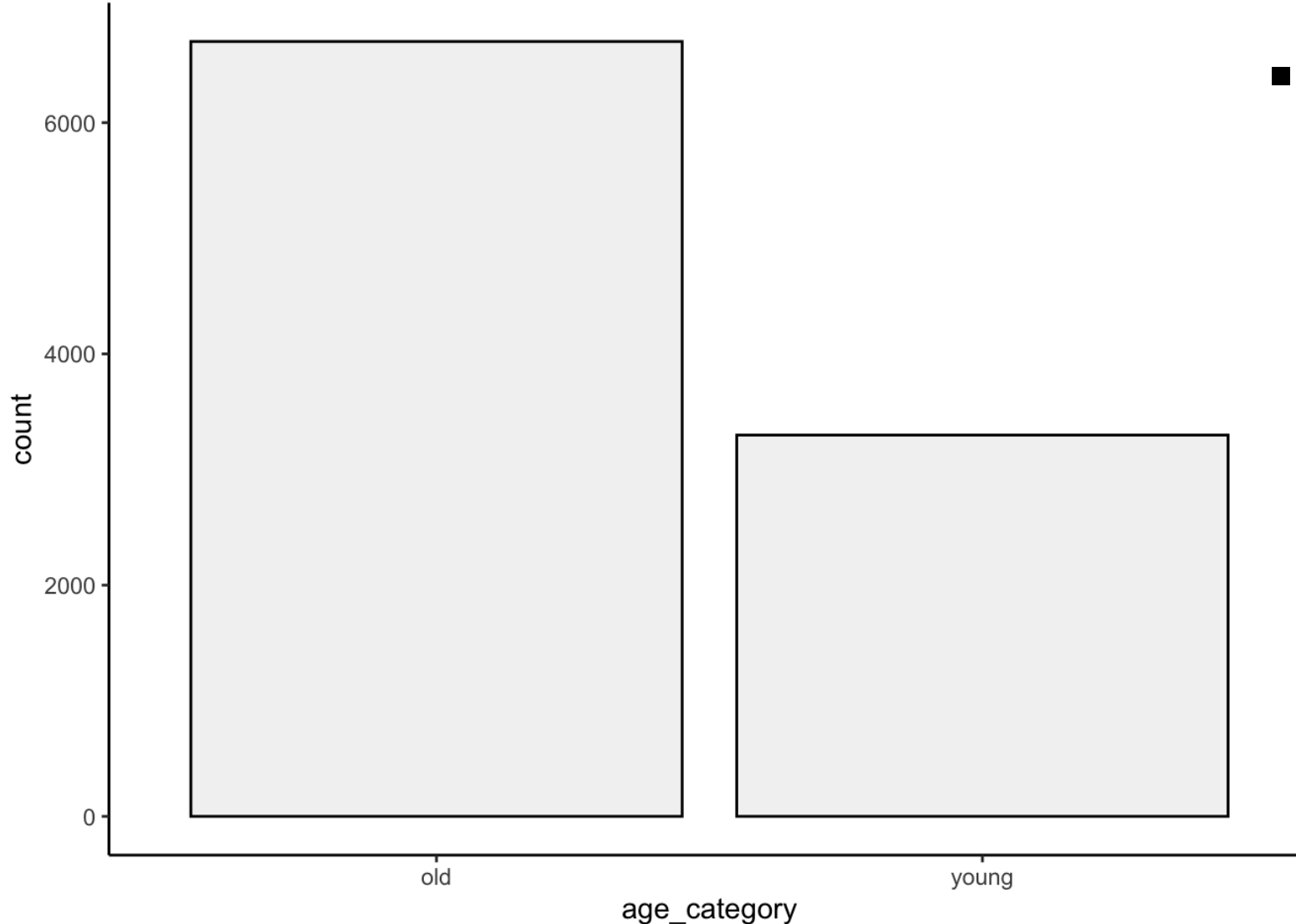


Count type distribution



- Poisson models
- Negative binomial models
- Zero inflated models

Bernoulli distribution



- Logistic regression
 - Probit regression
 - Tobit regression

Logistic regression

Logistic regression

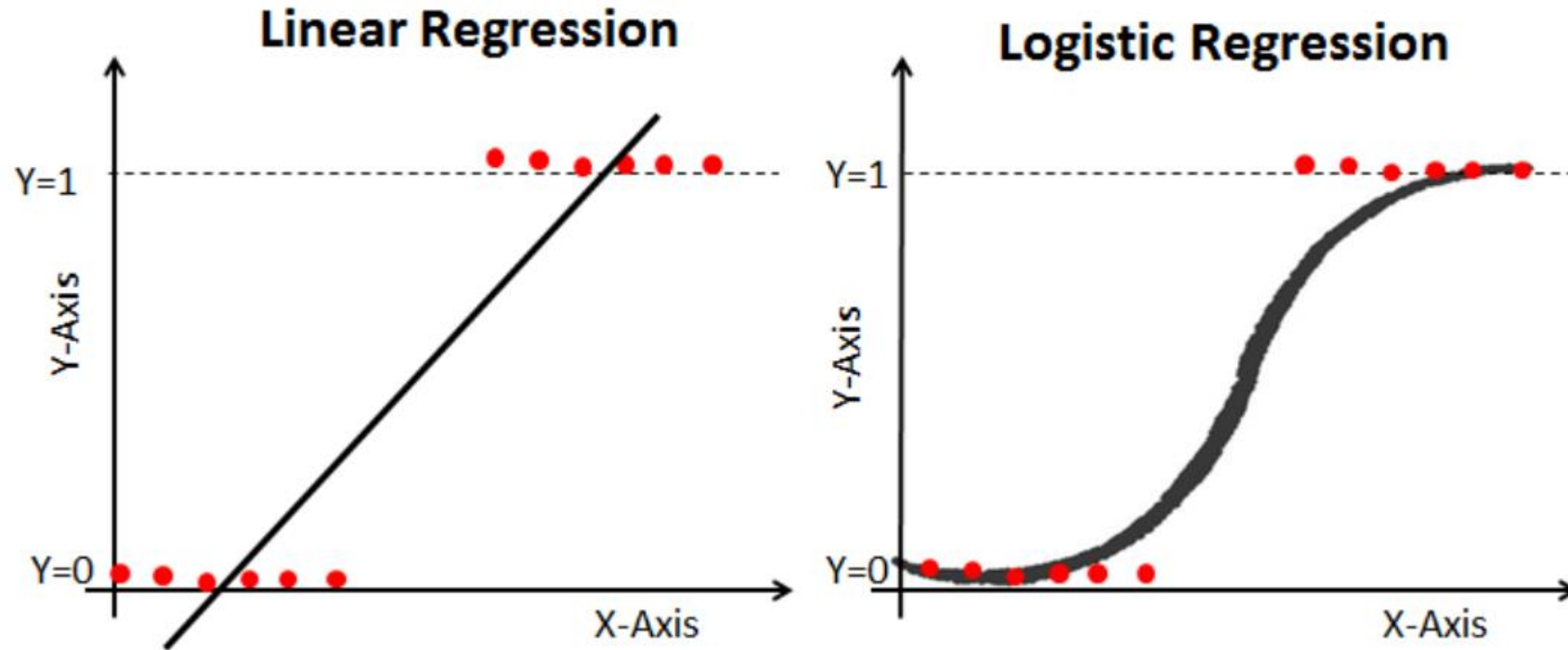
- Dichotomous outcome
- Modelling logit transformation of the probability of the outcome as a linear function of a set of predictor variables
 - a) Log odds of the outcome
- This transformation leads to the logistic model
- Can compute the odds of disease and odds ratio for disease that is associated with the presence of factor X (predictor)
- Can be extended to multiple predictors

Fitting a logistic model

- Set of parameter values that are most likely to have produced the observed data
 - a) Maximum likelihood (ML) estimates
 - b) In reality, these will be very, very small
 - c) Because of this (and because the estimation is easier) software packages usually work with the log likelihood
- Parameter estimate that gives you the maximum likelihood of having the binary outcome values 0/1

Typically for rare(ish) outcomes

- Distribution of 1/0 (cases/non cases) should be 5-15% for one and the inverse for the other.
- If the cases are too common logistic regression will overestimate the effect size
 - a) Probit regression
 - b) Tobit regression
- Software package side note
 - a) Make sure you are modelling the right level of your outcome variable
 - b) Some take 1 as the reference others take 0 as the reference by default



Likelihood ratios

- To determine overall significance of the model = likelihood ratio test
 - a) Compares the likelihood of the 'full' model (predictors added) with the 'null' model (no predictors)
 - Analogous to F-test in linear regression
 - b) For model comparison, the models must be nested (predictors in the simpler model must be a subset of those in the full model)
- To determine significance of individual predictors = Wald test

Model interpretation

- The coefficients in a logistic regression model represent the amount the logit of the probability of the outcome changes with a unit increase in the predictor
 - a) Not easy to interpret!
- Therefore – convert the coefficients to odds ratios
 - a) Exponentiation
- Continuous predictors: OR represents the factor by which the odds of disease are multiplied for 1-unit change in the predictor
- Categorical predictor: as with linear regression, convert to dummy variables – coefficient represents the effect of that level to baseline

Logistic regression: small group discussion

- We are interested in how variables, such as GRE score, grade point average and 'prestige' of the undergrad institution effect admission to graduate school
- Outcome: admit (binary variable: admit/don't admit)
- Predictors: gre, gpa (continuous variables) and rank (1 to 4; 1 is 'highest' prestige)
- Interpret the predictors
- Does this group of predictors reliably predict the dependent variable?
How do you know?

logit admit gre gpa i.rank

Iteration 0: log likelihood = -249.98826
 Iteration 1: log likelihood = -229.66446
 Iteration 2: log likelihood = -229.25955
 Iteration 3: log likelihood = -229.25875
 Iteration 4: log likelihood = -229.25875

Logistic regression	Number of obs	=	400
	LR chi2(5)	=	41.46
	Prob > chi2	=	0.0000
Log likelihood = -229.25875	Pseudo R2	=	0.0829

admit		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	

gre		.0022644	.001094	2.07	0.038	.0001202	.0044086
gpa		.8040377	.3318193	2.42	0.015	.1536838	1.454392
rank							
2		-.6754429	.3164897	-2.13	0.033	-1.295751	-.0551346
3		-1.340204	.3453064	-3.88	0.000	-2.016992	-.6634158
4		-1.551464	.4178316	-3.71	0.000	-2.370399	-.7325287
_cons		-3.989979	1.139951	-3.50	0.000	-6.224242	-1.755717

logit , or

Logistic regression

Number of obs = 400

LR chi2(5) = 41.46

Prob > chi2 = 0.0000

Log likelihood = -229.25875

Pseudo R2 = 0.0829

admit	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
gre	1.002267	.0010965	2.07	0.038	1.00012	1.004418
gpa	2.234545	.7414652	2.42	0.015	1.166122	4.281877
rank	1 1					
2	.5089309	.1610714	-2.13	0.033	.2736922	.9463578
3	.2617923	.0903986	-3.88	0.000	.1330551	.5150889
4	.2119375	.0885542	-3.71	0.000	.0934435	.4806919

Confounding and interaction

As with linear regression

- Confounding: ‘substantial’ (typically $>10\%$) change in coefficient
 - a) Statistical assessment – don’t forget to also include confounding variables from your DAG regardless of statistical assessment results
- Interaction: Cross-product term

Logistic regression assumptions

- Independence
 - a) Observations are independent from one another
- Linearity
 - a) Any predictor measured on a linear scale is assumed to have a linear relationship with the outcome
- Regression model diagnostics:
 - a) A whole topic in and of itself

Studying

- So many primers on logistic regression!
 - a) Harris JK. Primer on binary logistic regression. Fam Med Community Health. 2021 Dec;9(Suppl 1):e001290. <https://doi.org/10.1136%2Ffmch-2021-001290>
 - b) <https://github.com/jenineharris/logistic-regression-tutorial>

Goal of model building

- Best fit and parsimonious
- “Simplest model using the minimum number of parameters needed to explain a given phenomenon”
 - Raykov & Marcoulides, 1999

Model-building

- Focus here is on **statistical models** – but there are others
- History in epidemiology:
 - a) William Farr, 1840
- Decide on the goal of the model building
 - a) Predict vs causal
- Multivariable model building
 - a) Multivariate vs multivariable? <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3518362/>
 - b) Incorporate both statistical and subject matter knowledge
- Balance best fit with parsimony
 - a) Definition of ‘best fit’ and parsimonious will depend on the goal
 - b) For example, in machine learning, you may not want the most parsimonious

Objectives of model-building

- In general, regression modelling has one of two broad objectives:
 1. Best fitting model to predict future observations
 - Details of model may not matter (such as effect of specific predictors) but need to exclude any variable with a questionable relationship with the outcome
 2. Understand the relationship(s) (potentially causal) between one or more predictors and outcome of interest
 - Requires most precise coefficient estimates possible for variables of interest; careful attention to confounding and interaction
 - Most often the goal of epidemiological models

Model-building: Steps

1. Specify the maximum model to be considered i.e.) identify the outcome and full set of predictors
2. Specify the criterion/criteria to be used in selecting the variables to be included in the model
3. Specify the strategy for applying the criterion/criteria
4. Conduct the analyses
 1. Forward or backward step by step approach
 2. Stepwise forward or backward
 3. Best subset method.
5. Evaluate reliability of model chosen

Model building

- Is an **art** and a **science**
 - a) Both non-mathematical and mathematical
- Draw causal diagram
- Know your data, research question, study design, hypothesis, goal
- Iterative process
 - a) Fitting, assessing, comparing, diagnosing issues, fitting, assessing, comparing, diagnosing...
 - b) Potential interaction, confounding, violating model assumptions, 'best fit',
- Parsimony, best fit and interpretation

Model assumptions

Linear regression

- Validity and representativeness
- Additivity and linearity
- Independence of errors
- Equal variance of errors
- Normality of errors

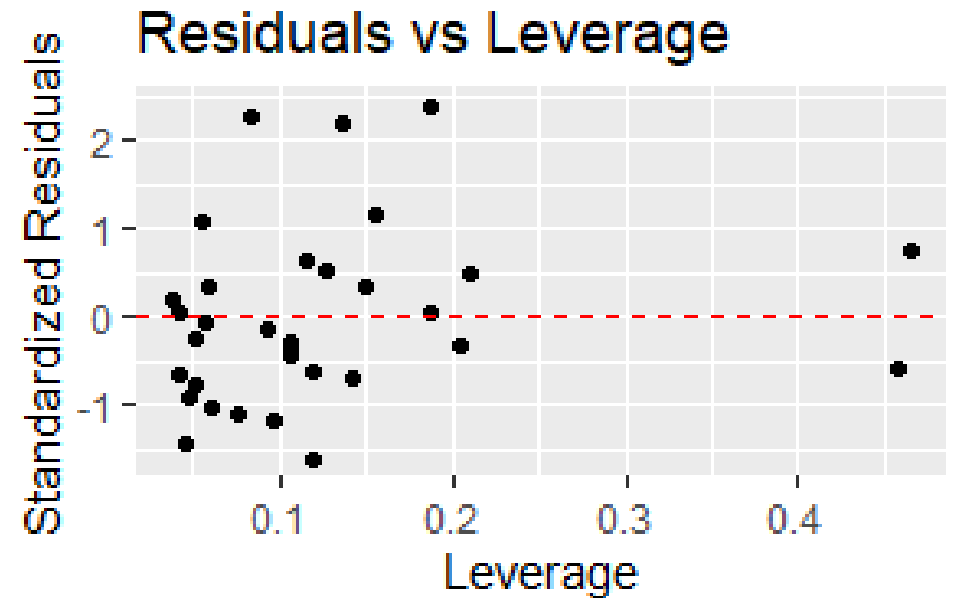
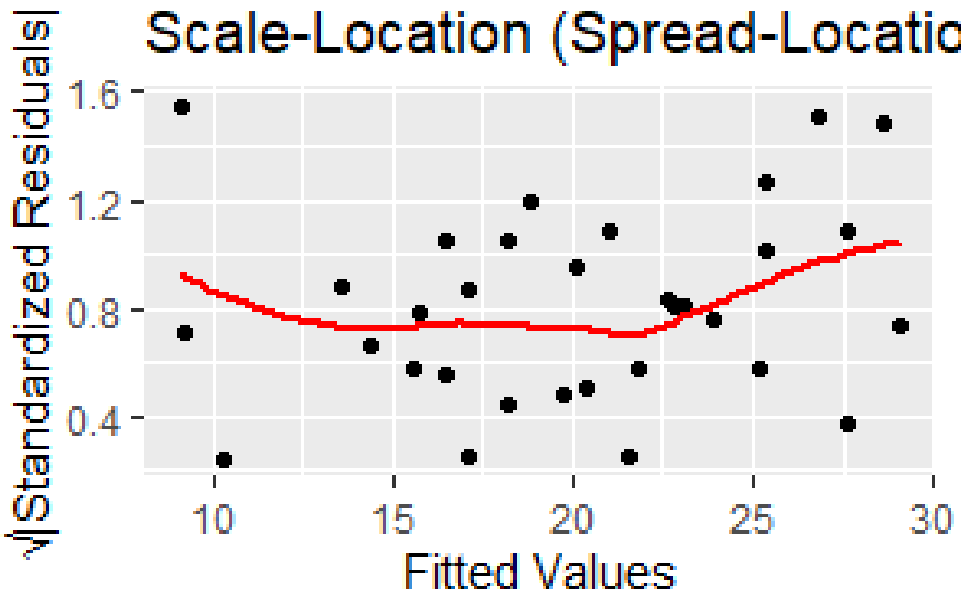
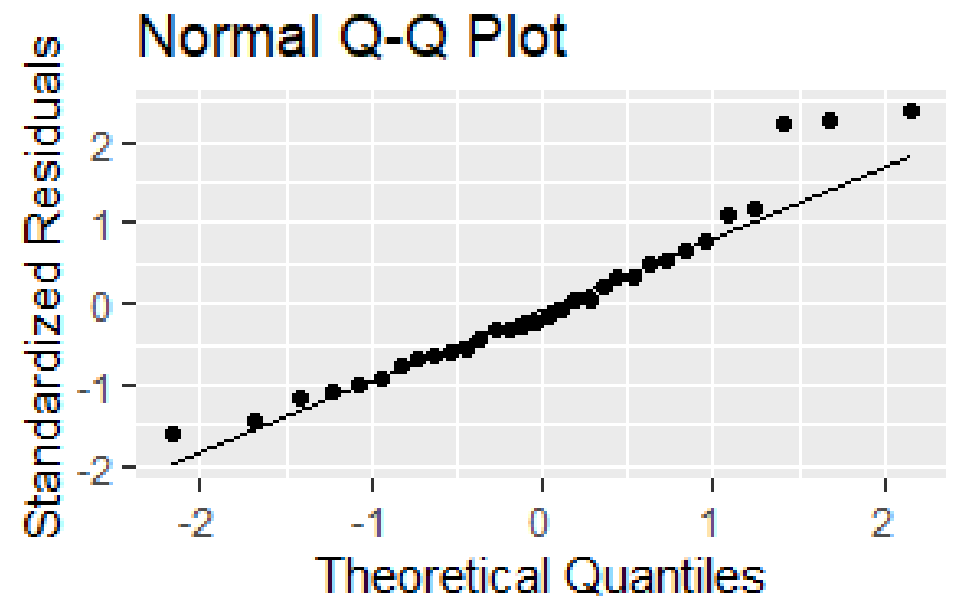
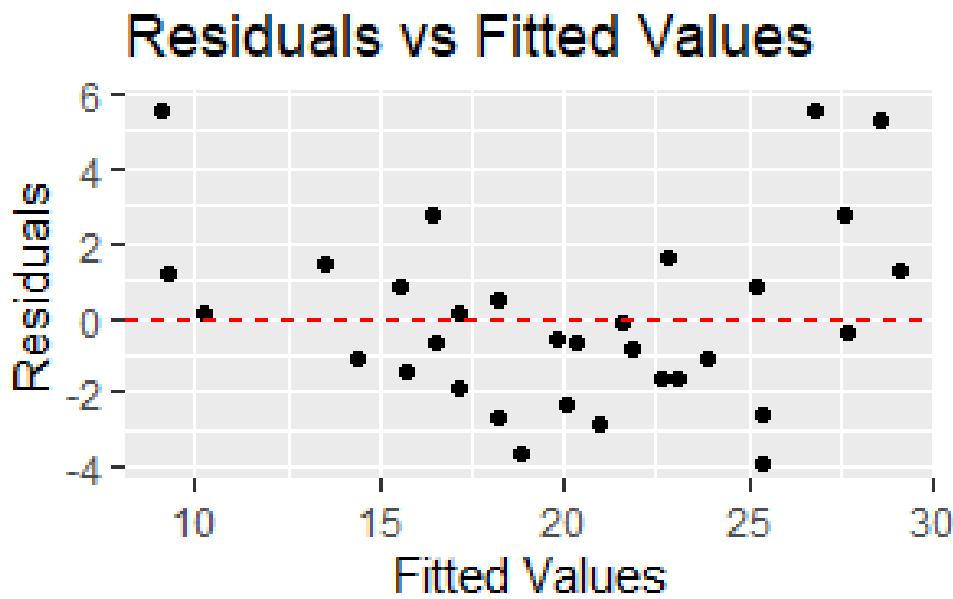
Logistic regression

- Validity and representativeness
- Additivity and linearity
- Independence of errors

How to assess?

To assess:

- Plot outcome vs fitted values
- Plot residuals vs fitted values
- Plot residuals vs continuous covariates
- Plot residuals vs time/spatial/cluster dependent covariates
- QQ plots of the residuals



Other

- Multicollinearity of predictors
 - a) Variance inflation factor
- Influence measures of particular observations/covariate patterns
- For binary and survival outcomes: calibration and discrimination

Paper Review

- Holzer KJ, Vaughn MG, Murugan V. Dog bite injuries in the USA: prevalence, correlates and recent trends. Injury Prevention 2019;25:187-190. <https://doi.org/10.1136/injuryprev-2018-042890>

- What is the outcome variables
 - a) How was it created?
- What are the predictor variables
 - a) Why were they included in the model
 - b) What are their associations with the outcome
- What is the study design?
- Is the outcome common or not?